Shayan Shakeri

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PROFESSIONAL SUMMARY

AI research engineer focused on LLM and agent evaluation, RL reward design, and the research infrastructure behind both. At BMO's AI Centre of Excellence, built a deterministic eval pipeline that uncovered systematic risk-downplaying bias in a GenAI tool serving $200B+ AUM. Co-founded Merlyn Labs: VLM judges that score rollouts into dense, hard-to-game RL rewards, an open-sourced flow-matching VLA integration for RLinf, and 8th place in Stanford's BEHAVIOR-1K Challenge.

CORE COMPETENCIES

LLM & Agent Evaluation | RL Reward Design | RL Environments | LLM Fine-tuning | Benchmark Analysis

WORK EXPERIENCE

BMO AI Centre of Excellence
Sep 2025 - Present

AI Research Engineer | Toronto, Canada

- Uncovered systematic bias to downplay investment risk in a GenAI tool serving **$200B+ AUM** wealth management
- Built a **deterministic agent eval pipeline** using hundreds of synthesized inputs to detect misaligned outputs at scale
- Building a **graph-based agentic system** answering complex multi-hop relational queries across bank client data

Merlyn Labs
Aug 2025 - Present

AI Researcher & Co-Founder | Toronto, Canada

- Developing **VLM judges** that score rollouts into dense, context-dependent **RL rewards that are difficult to game**
- Open-sourced flow-matching VLA integration for **RLinf** (open-source RL training infra), enabling RL on BEHAVIOR-1K in OmniGibson
- Showed π0.5 generalization failures on LIBERO-PRO are **recipe-induced overfitting**; proposed an alternative baseline
- Conservative finetuning doubled position-swap success (**21% to 42%**), showing spatial generalization over memorization

Epineuron Technologies
May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op | Mississauga, Canada

- Helped develop **PeriPulse**, an FDA Breakthrough-designated device now in multi-national clinical trials
- Optimized power draw via oscilloscope/power analyzer testing, achieving **900%** battery life improvement

RESEARCH & PROJECTS

Stanford BEHAVIOR-1K Challenge
8th Place | Standard Track — Sep 2025 - Nov 2025

- Identified **proprioceptive collapse** as critical VLA failure mode; 60% masking improved task success by up to **48%**
- Doubled manipulation success on **long-tail subtasks** by oversampling skill transitions via boundary resampling
- Trained on 10,000+ demonstrations across 22 tasks; published a technical report and further analysis on LessWrong

BardSong, AI-Powered D&D Storytelling Tool
Jun 2025 - Present

- Built end-to-end pipeline: Groq speech-to-text to Gemini scene extraction to image generation to narrated video
- **Fine-tuned LLM** on D&D sourcebooks for improved narrative coherence; currently in closed alpha with **23** DMs

EDUCATION

M.Eng, MIE - AI & Robotics | University of Toronto — Sep 2024 - Apr 2027 (anticipated)

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics

B.Eng, Engineering Physics (Co-op) | McMaster University — Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems

SKILLS & INTERESTS

ML & AI: Python, C/C++, PyTorch, JAX, RL, Imitation Learning, LLM Fine-tuning, Alignment Evaluation, Agents

Robotics & Hardware: VLA Models, OmniGibson, MuJoCo, Flow Matching, Sim2Real, ROS2, PCB Design, 3D Printing

Interests: Literature, Dungeons & Dragons, Volleyball, Twilight Imperium, Piano